

Brady A. Pettengill

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Computational and Data Sciences student at George Mason University who enjoys solving problems and building practical software, from data analytics to full-stack apps and cybersecurity challenges. Passionate about learning new tools and creating solutions that make technology smarter and more useful!

EDUCATION

Accelerated B.S./M.S. Computational and Data Sciences

Expected Dec 2026

Minor in Computer Science

George Mason University

Advanced Diploma, Herndon High School

June 2022

PROJECTS

Adaptive Cruise Control: Basic adaptive cruise control using computer vision with Python and OpenCV, communicating with the car via CAN bus and integrating with the existing cruise control through a PID controller.

Messaging App: Desktop chat platform with client-server functionality, written in Java using Swing and MySQL.

Discord Bot: Versatile bot with a range of features including music playback, weather fetching, image processing capabilities, a to-do list, and management tools. Developed in Python and later (partially) re-written in JavaScript.

Utility Website: A collection of various web apps built using the MERN stack (MongoDB, Express, ReactJS, NodeJS). Contains basic tools like a URL shortener and pastebin.

SKILLS

Languages: Python, C, C++, JavaScript, R, MATLAB, Java, SQL, Swift, Bash

Libraries: NumPy, Pandas, Scikit-learn, OpenCV, MySQL, MongoDB, NodeJS, Express

Tools/Platforms: Git, Linux, Docker, NGINX, Cloudflare, GCP, Wireshark, Ghidra, GDB, Nmap

Soft Skills: Team collaboration, problem-solving, adaptability, technical communication, leadership

EXPERIENCE

George Mason Cyber Team

Sep. 2023 – Present

CTF Competitor

Fairfax, VA

- Compete in NCL and collegiate CTFs, focusing on web, crypto, and reverse-engineering challenges.
- Apply tools such as Burp Suite, Wireshark, Nmap, Ghidra, and GDB to identify and exploit vulnerabilities.
- Collaborate with teammates to solve real-world-style cybersecurity problems under tight time constraints.

State Farm Insurance

May 2024 – Sep. 2025

Digital Marketing Coordinator

Vienna, VA

- Created and managed digital advertising campaigns, building an online presence with the goal of driving sales.

Nando's PERi-PERi

Dec. 2021 – May 2024

Restaurant Manager

Reston, VA

- Oversaw financial performance, inventory management, marketing and outreach, and hiring/scheduling staff.

FIRST Robotics FTC Team

Aug 2021 – May 2022

Software Team Lead

Reston, VA

- Co-founded the non-school-affiliated FTC robotics team "Independent Robotics."

- Developed software for our robot, including both manual control and autonomous operation.

Herndon High School AFA CyberPatriot Team

Sep. 2020 – June 2022

Founder and President

Herndon, VA

- Founded HHS CyberPatriot team "CyberHornet," reaching national semifinals in both years of competition.
- Focused on the development of our Linux sub-team, introducing members to the use of Linux-based systems.
- Worked extensively with bash scripting to automate standard competition procedures.